PHYSICS

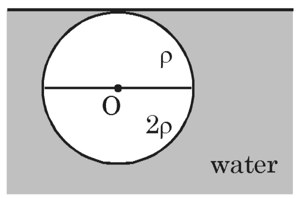
TARGET : JEE-ADVANCED-2024

CAPS-7$^+$

1. If the β-rays spectrum is represented by $n \propto (E_{max} - E)^2 \times \sqrt{E}$ where n is the number of β particles emitted with energy E and E_{max} is energy released in β decay, then maximum number of neutrinos would come out with energy equal to (approximately) :–

(A) $\frac{E_{max}}{5}$ (B) $\frac{4E_{max}}{5}$ (C) E_{max} (D) 0

2. A spherical ball of radius R is made by joining two hemispherical parts. The two parts have density ρ and 2ρ. When placed in a water tank, the ball floats while remaining completely submerged. Neglect viscous forces. Mark the incorrect statement:



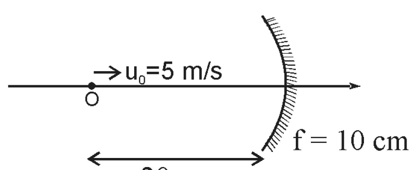
(A) Density of water is ρ_0, then $\rho = \frac{2\rho_0}{3}$

(B) Position of centre of mass of ball is at distance of $\frac{R}{8}$ from O.

(C) Moment of inertia of ball about COM of ball perpendicular to the plane of paper is $\frac{123}{160}\rho\pi R^5$

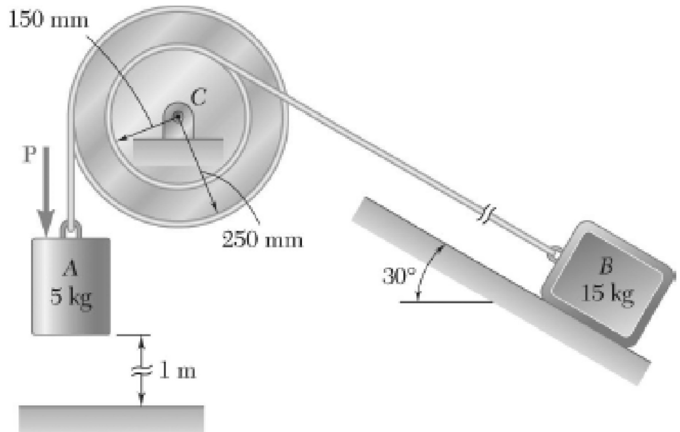
(D) Time period of small angular oscillations of the ball about its equilibrium position is given by $2\pi\sqrt{\frac{113R}{40g}}$

3. An object moves with a uniform velocity u_0 along the axis of a concave spherical mirror. Consider the instant shown in diagram, object is moving with u_0 = 5 m/s and f = –10 cm. If object is at the centre of curvature at this instant then the magnitude of acceleration of image at this instant is a m/s^2. Find a.



(A) 10 (B) 5 (C) 25 (D) None of these

4. The double pulley shown has a mass of 15 kg and a centroidal radius of gyration of 160 mm. Cylinder A and block B are attached to cords that are wrapped on the pulleys as shown. The coefficient of kinetic friction between block B and the surface is 0.2. Knowing that the system is at rest in the position shown when a constant force P = 200 N is applied to cylinder A, the velocity of cylinder A as it strikes the ground is



(A) 47.9 m/s (B) 4.79 m/s (C) 0.479 m/s (D) None of these

5. An X-ray tube operates at 50 kV. Consider that at each collision, an electron converts 50% of its energy into photons and 10% energy would be dissipated as thermal energy due to the collision, then the wavelength of emitted photons during 2^{nd} collision is [Take he = 1242 eV-nm]

(A) 1.242 nm (B) 1.242 Å (C) 4.968 nm (D) 4.968 Å

6. A single conservative force acts on a 1 kg particle that moves along x-axis. The potential energy of the particle varies with x as $U = 20 + (x-2)^2$, here U is in joules and x is in meters. When the particle is at x = 5 m, its kinetic energy is 20 J, then which of the following is/are correct?

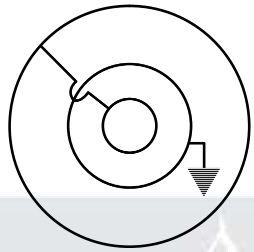
(A) Mechanical energy of particle is 49 J.

(B) Least and greatest value of x between which particle can move is $(2-\sqrt{29})$m and $(2+\sqrt{29})$m respectively.

(C) Maximum kinetic energy of the particle is 29 J.

(D) At x =2, the body is in equilibrium.

7. There are three concentric spherical shells with radius 4R, 6R and 12R. Initially no charge is present on the shells. Charge of 3Q is given to outermost sphere. After this inner and outer-most shells are connected by using a thin conducting wire which passes through a small insulated hole on middle sphere. The middle shell is grounded. Choose the incorrect options from the following :



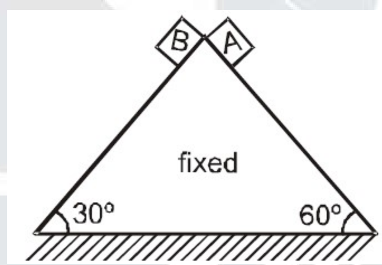
(A) Charge on inner surface of middle sphere is –Q

(B) Capacitance of system is 8 $\pi\varepsilon_0 R$

(C) Charge on outer surface of outer sphere is 2Q

(D) Capacitance of system is 48 $\pi\varepsilon_0 R$

8. Small blocks A and B are simultaneously released from apex of a smooth wedge as shown in figure, select correct alterative(s) :



(A) Relative acceleration of block B with respect to block A is zero

(B) Magnitude of relative acceleration of block B with respect to block A is g initially

(C) Speed of block A and B will be same at the bottom of inclined plane

(D) The time taken A and B to reach the bottom of inclined plane will be same.

9. The spherical planets have the same mass but densities in the ratio 1: 8. For these planets, the

(A) acceleration due to gravity will be in the ratio 4: 1

(B) acceleration due to gravity will be in the ratio 1: 4

(C) escape velocities from their surfaces will be in the ratio $\sqrt{2}:1$

(D) escape velocities from their surfaces will be in the ratio $1:\sqrt{2}$

10. Consider a vernier caliper. Least count of main scale is 1mm. There are 20 equispaced marks on vernier scale. When nothing is put between the jaws, zero of main scale lies between zero and first division of vernier scale and and 15th division of vernier exactly coincide with one of the main scale division. When zero of vernier is matched with zero of main scale, 20th division of vernier exactly coincides with 25^{th} division of main scale. To measure the side of a cuboid, it is kept between the jaws of vernier caliper, it is now observed that 18th division of vernier exactly coincides with 32nd division of main scale. Choose the correct option(s).

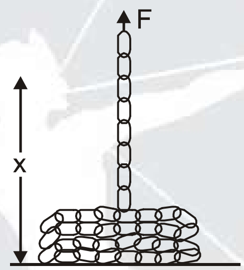
(A) Least count of vernier scale is 0.25 mm

(B) Zero error is -0.25 mm

(C) Side of cuboid is 10.25 mm

(D) Side of cuboid is 9.25 mm

11. A chain of length L and mass per unit length λ, lies on a horizontal surface. If one end of the chain is lifted vertically with a constant velocity v by a force F. Identify correct statements for $(x < L)$ given Situation. (Note :- Height of heap is negligible.)



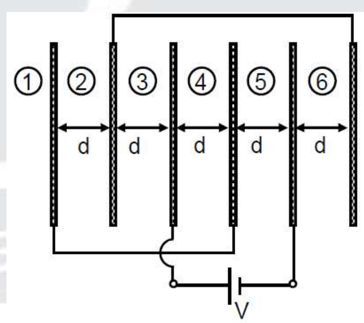
(A) $F = \lambda(gx + v^2)$

(B) Work done by force F = $\frac{\lambda g x^2}{2} + \lambda v^2 x$

(C) Energy loss in this time is $\frac{\lambda x v^2}{2}$

(D) There is no energy loss in the process

12. Consider an arrangement of six large parallel plates as shown in the figure, each of plate area A. Initially when battery was not connected all plates were uncharged. The emf of the battery is V. Choose the correct option(s) after connecting the battery (in steady state):



(A) Total work done by battery is $\frac{\varepsilon_0 A}{d} V^2$

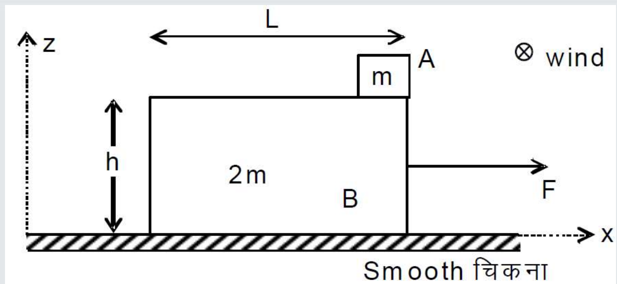
(B) Electric field between plates 3 and 4 is $\frac{V}{2d}$

(C) Potential difference between plate 2 and 5 is $\frac{V}{2}$

(D) Potential difference between plate 1 and 5 is $\frac{V}{2}$

COMPREHENSION

Two blocks A and B of masses m and 2m are initially at rest. Length of block B is L and the block A is placed at the right end corner of block B and the friction coefficient between them is μ = 1/2. At t = 0 a constant force $F = \frac{5mg}{2}$ begins to act on block B towards right. Just when the block A leaves B, wind begins to blow along y - direction which exerts a constant force $\frac{mg}{2}$ on A. Assume the size of block A is small compared to B and neglect any rotational effects and toppling of block B. (Given h = 1/2 m, L = 1m and g = 10 m/s^2)



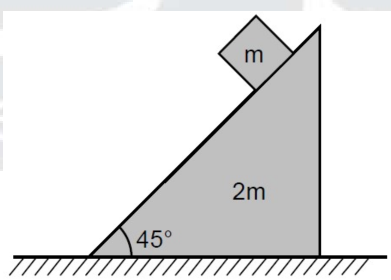
13. Find ratio of the displacements of block A along x and y directions S_x/S_y after the time block A leaves the surface of B till the time it reaches ground

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) 4 (D) $\sqrt{\frac{8}{5}}$

14. The magnitude of relative acceleration of A with respect to B (in m/s^2) just after the block A leaves B is (assume wind does not affects motion of B)

(A) $\sqrt{10}\,g$ (B) $\frac{\sqrt{29}\,g}{4}$ (C) $\frac{9\sqrt{5}}{4}$ (D) $\frac{3\sqrt{5}}{4}g$

15. A wedge of mass 2m and a cube of mass m are shown in figure. Between cube and wedge, there is no friction. The minimum coefficient of friction between wedge and ground so that wedge do not move is x/10 find the value of x.



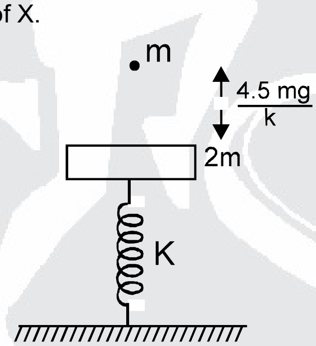
16. A point sound source is situated in a medium of bulk modulus 1.6 × 10^5 N/m^2. An observer standing at a distance 10 m from the source, writes down the equation for the wave as y = A sin (15 πx – 6000 πt); y and x are in metre and t is in second. The maximum pressure amplitude received to the observer's ear is 24π Pa, The power of sound source is x times π^3 watt. The value of x is ____ .

17. When a man moves down the inclined plane with a constant speed 5 m/s which makes an angle of 37° with the horizontal, he finds that the rain is falling vertically downwards, when he moves up the same inclined plane with the same speed, he finds that the rain makes an angle of $\tan^{-1}\left(\frac{7}{8}\right)$ with the horizontal. If the speed of the rain is $\sqrt{R}\,m/s$. Find R.

18. The potential energy of a 1 kg particle free to move along the x-axis is given by U = $\left(\frac{x^4}{4}-\frac{x^2}{2}\right)$ J. The total mechanical energy of the particle is 2J. If the maximum speed is $\frac{x}{10}$ m/s. Calculate 'x'.

19. A siren creates a sound level of 60 dB at a location 500 m from the speaker. The siren is powered by a battery that delivers a total energy of 1.0 kJ. Assume that the efficiency of siren is 30%. The time for which siren can sound is (x ± 1) sec. The value of x is

20. In the figure shown a plate of mass 60gm is at rest and in equilibrium. A particle of mass m = 30gm is released from height $\frac{4.5mg}{k}$ from the plate. The particle sticks to the plate. Neglecting the duration of collision find time from the collision of the particle and the plate to the moment when the spring has maximum compression. Spring has force constant 100 N/m. Calculate value of time in the form Xπ ms (millisecond) and fill value of X.



ANSWER KEY

1.	(B)	**2.**	(D)	**3.**	(B)	**4.**	(B)	**5.**	(B)
6.	(ABCD)	**7.**	(BCD)	**8.**	(BC)	**9.**	(BD)	**10.**	(AC)
11.	(ABC)	**12.**	(ABCD)	**13.**	(C)	**14.**	(D)	**15.**	x = 2
16.	288	**17.**	32	**18.**	21.21	**19.**	95	**20.**	20.00